

Cayley-Hamilton theorem

Key Points

The Cayley-Hamilton theorem states that:

"A matrix satisfies its own Characteristic equation"

That is replacing λ with M solves the new matrix equation (with identities and zero matrix)

This matrix equation can then be manipulated to find powers/inverses of M in terms of other powers of M

Basic Skills

Q1. What is meant by the characteristic equation

Q2. Write down the characteristic equation of

a) $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ b) $\begin{pmatrix} -1 & 3 \\ 1 & 5 \end{pmatrix}$ c) $\begin{pmatrix} 5 & 1 & 4 \\ 0 & 4 & -1 \\ 2 & 6 & -2 \end{pmatrix}$

Q3. Rearrange the equation $M^2 - 6M + 3I_2 = 0$ to find

- a) M^2 in terms of M, I_2
- b) M^5 in terms of $M^4, \dots, M, I_2$
- c) M^{-1} in terms of M, I_2

Competency Skills

Q4. For the matrix $M = \begin{pmatrix} 6 & 5 \\ -1 & 4 \end{pmatrix}$

- a) Show that $M^2 - 10M + 29I_2 = 0$
- b) By post-multiplying by M^{-1} show that $M^{-1} = \frac{1}{29}(10I_2 - M)$ and hence explicitly state M^{-1}

Q5. For the matrix $M = \begin{pmatrix} 0 & 2 \\ 1 & -2 \end{pmatrix}$ use the Cayley-Hamilton theorem to find the inverse M^{-1}

Q6. For the matrix $M = \begin{pmatrix} 6 & 5 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & -2 \end{pmatrix}$

- a) Find M^3 by using the Cayley-Hamilton theorem
- b) Find M^{-1} by using the Cayley-Hamilton theorem

Advanced Skills

Q7. For the matrix $M = \begin{pmatrix} 7 & -1 \\ 8 & 2 \end{pmatrix}$

- a) Show that $M^2 - 9M + 22I_2 = 0$
- b) Write a recursive formula for M^n in terms of M^{n-1}, M^{n-2}

Q8. Given that a square matrix of order 2 called M has eigenvalues $\lambda = -2, 3$

- a) Show that $M^2 - M = 6I_2$
- b) Decide if M is a unique matrix

Q9. Prove the Cayley-Hamilton theorem for 2×2 matrices

Extension

Q10. The Cayley-Hamilton theorem can be used to find matrix functions such as $f(M) = e^{Mt}$. Research how this is possible (Wikipedia has a decent explanation if you fail to find anything else)